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Predation on early juvenile spiny lobsters *Panulirus argus* (Latreille): influence of size and shelter

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Abstract: Juvenile spiny lobsters *Panulirus argus* (Latreille) from three behaviorally and ecologically distinct ontogenetic groups (algal, 5–15 mm carapace length; transitional, 16–25 mm CL; and post-algal, 26–35 mm CL) were tethered in their characteristic shelters and on open substratum to evaluate size related differences in predation risk. Field experiments performed at two sites near Long Key, Florida Bay nursery habitat indicate that juveniles attain a partial size refuge from a suite of abundant algal lobster predators at about the time they emerge from settling habitat. Algal lobsters experience significantly decreased mortality by sheltering at night, thereby attaining a survival rate comparable to that of larger, older juveniles that forage nocturnally in the open. Diver surveying and limited net sampling revealed an array of lobster predators including octopus, portunid crabs, bonnethead sharks, nurse sharks, sting rays, gray snapper and toadfish, as well as general crustacean predators including bonefish and permit. High relative mortality of the smallest juveniles suggests that predation on the algal and early transitional phases is a potential bottleneck to population recruitment.

Key words: *Panulirus argus*; Predation; Tethering experiment; Size refuge

Introduction

Predation is a major force shaping aquatic community structure (Connell, 1961a,b, 1972; Paine, 1974, 1976; Lubchenco, 1978; Choat, 1982; Mittelbach, 1988; Butler, 1989) and extensively influencing prey morphology, life history, and behavior (Sih, 1985, 1987). The effects of predation are realized at both the community level and at different age-related size ranges within a single species; predation rate usually decreases markedly at older, larger life stages (Paine, 1976). Tethering experiments performed with a variety of small crustaceans show that remaining in thickly vegetated habitats significantly reduces predation as compared to more open substrata (Heck & Thoman, 1981; Herrnkind & Butler, 1986; Wilson et al., 1987). At the population level, high predation on early juveniles reduces the number of recruits available for subsequent life stages, thereby presenting a potential recruitment bottleneck to the population (Underwood & Denley, 1984; Caddy, 1986; Gaines & Roughgarden, 1987; Menge & Sutherland, 1987).

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Ample evidence exists that clawed and spiny lobster juveniles may be strongly influenced by predation in various ways (Marx & Herrnkind, 1985b; Herrnkind & Butler, 1986; Barshaw & Lavalli, 1988; Howard, 1988). Caribbean spiny lobsters *Panulirus argus* (Latreille) (Crustacea: Palinuridae) exhibit several distinct ontogenetic shifts in sheltering and foraging behaviour. In Florida Bay, free-swimming spiny lobster pueruli (postlarvae) settle into structurally complex vegetation, especially Rhodophyte algae, genus *Laurencia* (Marx & Herrnkind, 1985a; Herrnkind & Butler, 1986; Marx, 1986). Within days, the settled pueruli take on cryptic patterns of coloration, metamorphosing into the first benthic juvenile stage after about one week during summer (Butler & Herrnkind, in press). Early juveniles continue to exhibit light-dark banding and striping on appendages and body providing them with outline-disruptive color patterns when in algae or other complex visual backgrounds (Marler & Hamilton, 1966; pers. obs). The juveniles remain for several months within vegetation where they forage on small epifaunal invertebrates (Marx & Herrnkind, 1985b; Herrnkind et al., 1988). Camouflage pigmentation, cryptic and solitary behaviour, as well as the physical shelter of algae (Herrnkind & Butler, 1986) provide early juveniles substantial refuge from predation. After several months, the juveniles make a behavioral and ecological transition, while concomitantly taking on adult color patterns (Andree, 1981; Marx & Herrnkind, 1985a). Juveniles at this phase shift from sheltering in algae to daytime residence in small crevices, or other cover, located close to vegetation where they feed nocturnally (Andree, 1981; pers. obs.). After several months, juveniles are most commonly found by day under sponges, soft corals, sea grass ledges, within solution holes and other cover. They forage nocturnally in adjacent open habitat for increasing durations and over greater distances as they grow (Davis, 1971; Herrnkind, 1980; Andree, 1981). At this time they also become increasingly social and commonly aggregate within their diurnal shelter (Kanciruk, 1980; Marx & Herrnkind, 1985a). Based on the sheltering data of Marx & Herrnkind (1985a) and Andree (1981), we refer to the three ontogenetic phases as algal, transitional and post-algal.

These behavioral shifts hypothetically reflect adjustments to changing predation risk. Algal juveniles, by remaining within a structurally complex habitat, potentially avoid very high levels of predation. Only after transitional juveniles attain a size refuge from the suite of algal lobster predators would they be expected to emerge from protective cover and forage nocturnally in the open. As they grow larger, post-algal juveniles should experience continually decreasing predation risk permitting longer exposure while foraging (Andree, 1981). Recent work has shown that lobster size in relation to shelter features influences predation mortality on larger juvenile spiny lobster (Eggleston et al., 1990).

Field experiments on mobile marine invertebrate prey generally require limiting an animal's range of movement using techniques such as predator exclusion or inclusion caging (Virnstein, 1977, 1979; Peterson, 1979, 1982), or tethering (Heck & Thoman, 1981; Schulman, 1985; Herrnkind & Butler, 1986; Heck & Wilson, 1987; Wilson et al., 1987, in press). Such techniques are effective for comparing predation levels, although they cannot measure absolute predation because of the artificial constraints on either the prey or predator (Heck & Thoman, 1981; Herrnkind & Butler, 1986; Wilson et al., 1987; Barshaw & Able, 1990).

The present field study used field tethering to evaluate relative predation mortality on a range of sizes of early juvenile Caribbean spiny lobster with respect to shelter and exposure during the algal, transitional and post-algal phases. In addition, observation and limited netting were performed to qualitatively evaluate the range of juvenile lobster predators in the study habitat.

Materials and Methods

FIELD PROCEDURES

Field studies were performed in a typical Florida Bay spiny lobster nursery habitat near Long Key, Florida, where depth averaged ≈ 2 m. The area contained numerous juvenile lobsters from algal phase to subadult sizes (Herrnkind & Butler, 1986; pers. obs.). Two sites were chosen for tethering experiments (Fig. 1). The KML site (N 24° 48′ 73″, W 80° 48′ 48″) was located 50 m offshore of the Keys Marine Laboratory. The second site, KOA (N 24° 50′ 55″, W 80° 47′ 63″), was ≈ 3 km to the northeast of KML and 50 m offshore of Fiesta Key. Tethering experiments were performed during 1988 at KML in June and July and at KOA in July and August. Transects of each location indicated they had large areas of both sandy substratum and benthic algae (*Laurencia* spp.) interspersed with patches of sea grass (*Thalassia testudinum* [Koenig].), sponges and soft corals (Table I).

Algal juveniles, tank-reared from plankton-netted postlarvae, were fed flaked fish food and natural prey (crushed and live snails) until they attained the desired size. We

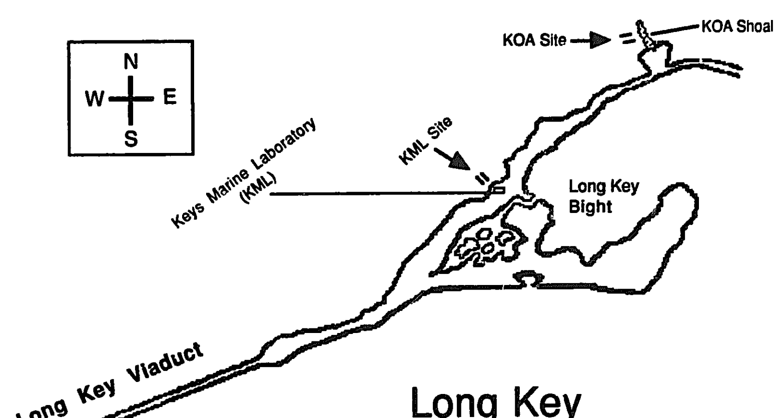

Fig. 1. Representation of study sites in relation to Long Key.

TABLE I

Average percent coverage of substratum at KML and KOA study sites. Data are derived from averages of distances along two 50-m transects covered by identified habitat ± 1 SD. Habitat types include open bottom (sand covered limestone), sea grass (predominantly *Thalassia* spp., algae (*Laurencia* spp. or *Dictyota* spp.), sponge (includes: black chimney sponge *Pellina carbonaria*, stinking vase sponge *Ircinia strobilina*, red finger sponge *Amphimedon compressa*, and common loggerhead sponge *Spheciospongia vesparium*) and seawhips (predominantly, *Leptogorgia* spp. and *Pterogorgia* spp.).

	Open Substratum	Sea grass	Algae	Sponge	Sea whip
KML	38.6 ± 2.0	17.4 ± 5.0	34.1 ± 7.2	4.8 ± 4.0	1.8 ± 0.6
KOA	39.0 ± 5.9	2.2 ± 1.1	50.7 ± 8.0	6.6 ± 4.4	6.1 ± 0.7

also provided small clumps of natural and artificial algae which they readily accepted as shelter. Larger juveniles, including transitional and post-algal lobsters, were collected from nearby natural habitats by scuba divers using nets. Juveniles were maintained outdoors in flow-through seawater tanks, provided perforated concrete blocks as shelter, and fed live gastropods (*Cerithium* sp.) ad libitum.

Field tethering was used to test for differences in the relative predation rates on juveniles from 5–35 mm carapace length (CL; measured from the carapace margin between the rostral horns to the posterior edge of the cephalothorax), using techniques patterned after Herrnkind & Butler (1986). Hard-shelled, intermolt juveniles were removed from the holding tanks, measured and assigned to separate seawater containers. Tethering involved gently drying the cephalothorax and attaching a 30-cm length of clear monofilament fishing line. Monofilament line of 3.7 kg test was used for lobsters 5–15 mm CL while line of 5.5 kg test was used for lobsters > 15 mm CL. The lobster's abdomen was first inserted through an overhand loop tied in the line which was then drawn snug around the cephalothorax between the fourth and fifth walking legs. The loop was secured by a droplet of cyanoacrylate gel glue applied to the knot and dorsal exoskeleton. The free end of the tether was then tied to a 10-cm long galvanized nail which was later hammered into the limestone substratum. Preliminary observations showed that none of the lobsters in any size class escaped the tethers over an 86-h period and lobsters did not behave erratically while on tethers. Larger juveniles occasionally pulled against the tether, but mostly sat quiescently or moved about slowly as described by Herrnkind & Butler (1986).

Juvenile lobsters were subdivided into three size classes approximating the three different behavioral phases: 5–15 mm CL (algal stage); 16–25 mm CL (transitional stage lobsters); 26–35 mm CL (post-algal stage juveniles). Trials at KML used eight lobsters from each size class, while eight algal lobsters and 12 each from the larger size classes were used at KOA. Lobsters were randomly assigned to predetermined coordinates on the transect grid, then tethered and secured until use in appropriate prenumbered positions in 12-l styrofoam boxes filled with aerated seawater. Tethered lobsters were held for 3–4 h prior to field experiments; dead or moribund lobsters were discarded and replaced by juveniles of equal size.

Each site contained two 60-m transect lines (23-kg test monofilament) strung between concrete anchors marked with numbered fluorescent surface buoys. The transects were laid out 15 m apart ≈ 50 m offshore and perpendicular to the shoreline. Several hours before each trial, equal numbers of open and sheltered sites were chosen for each size class and marked by numbered plastic flags on stiff plastic-coated wire. Open sites of naturally exposed substratum were chosen to be at least 80 cm in diameter so lobsters could not reach cover. Shelters were selected according to the characteristic habitat of each size group. Algal lobsters were placed solely in *Laurencia* while post-algal juveniles were placed around the undercut bases of sponges, foliose sea whips, or in small rock crevices. Transitional lobsters were evenly divided between algae and the other shelters because such juveniles normally occupy the complete range of cover. Numbered tethered sites along the transects were separated by 3–4 m and placed up to 2.5 m to either side of the transect line.

Each of five experimental runs at both sites was begun just after sunset by placing the tethered lobsters at the flagged locations. Transects were surveyed after 10 h (sunrise) and 24 h (sunset). Predation was determined by a cut or broken tether or an intact loop with carapace fragments attached (Herrnkind & Butler, 1986). The location of each predation event was recorded at the time of each survey. Prior to each survey, we swam along the transects and recorded the number and estimated size of observed predatory fish and invertebrates.

ANALYSIS

Initially, a step-wise logistic regression (BMDP; program LR; 1987 version) was performed on individual, categorically arranged predation data to determine if the percent predation values differed by site and experimental run (see Wiesberg, 1980). Replicate and site effects were determined to be nonsignificant by the logistic regression method. Data from the experimental sites were therefore pooled and the percent predation for replicates was calculated for each size class during both 10-h (night) and 24-h surveys. These combined data were analysed by two-way ANOVA (SYSTAT; MGLH program; version 5.0) with class and habitat as independent variables (Sokal & Rohlf, 1969). Prior to analysis, data were arcsine-transformed to satisfy the assumption of homogeneity of variances between groups (see Bartlett, 1947, cited in Snedecor & Cochran, 1980). Relevant combinations of size and shelter (variables) effects (on percent predation) were analyzed a posteriori as block means using paired Bonferroni *t* tests of independent variable (categorical) combinations (SYSTAT Statistical Software, Evanston, Illinois).

PREDATOR SAMPLING

In addition to visual observations, large, transient, and diurnally cryptic predators were sampled by trammel nets. In August 1988, following the last field experiment, two 25-m nets were deployed in sequence between transects. Nets were set on three successive days at 1730 then checked at 2030 and again at midnight to account for both crepuscular (late afternoon and dusk) and nocturnal predators. Whole small fish (< 0.7 m) and the excised gut from large fish (> 0.7 m) were preserved in a 10% buffered formalin/filtered seawater mixture. In the interest of conservation, we purposely limited the netting effort, as it was intended to be primarily descriptive.

RESULTS

24-h predator levels were plotted as percent predation by 3 mm CL increments for all lobsters tethered in the open vs. those in shelter (Fig. 2). Visual inspection of trends shows that initially high mortality decreased rapidly from 5–7 mm CL to 17–19 mm CL after which it remained relatively constant up to 35 mm CL for animals tethered in the open. Lobsters tethered within shelter also experienced declining predation with increasing size, although the change was marked only between 5–7 and 8–10 mm CL juveniles and seemed to level out soon thereafter. The sheltered 5–7 mm CL group suffered obviously greater predation than other sheltered size classes.

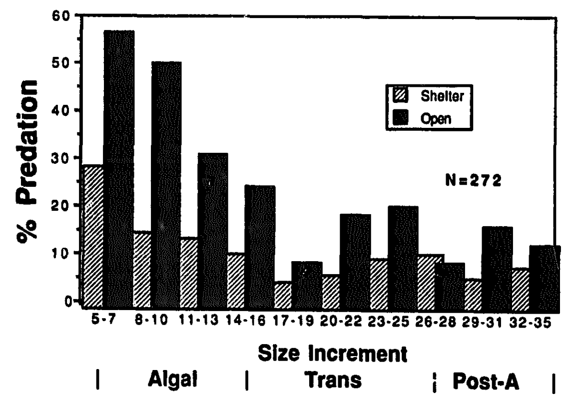

Fig. 2. Total percent predation for all tethered juvenile lobsters divided into 3 mm CL size increments: shelter vs. open. Data represent total predation on each size increment for all combined experimental replicates obtained from 24-h tethering experiments within shelter and on open substratum.

Predation from each ontogenetic class was compared statistically for open vs. sheltered lobsters in the 10-h nighttime and full 24-h periods (Fig. 3). Results of a two-factor ANOVA (Table IIA) applied to nighttime predation data, using size and presence or absence of shelter as factors, show significant effects for both size ($p = 0.0001$) and shelter ($p < 0.01$). Interaction between these two factors was also significant ($p < 0.01$) indicating that different sized lobsters experienced unequal predation in shelter. Nighttime predation levels in the open declined rapidly from the algal to the transitional phase followed by a less dramatic reduction from transitional to postalgal sizes. Nightly mortality of algal lobsters in the open was significantly higher

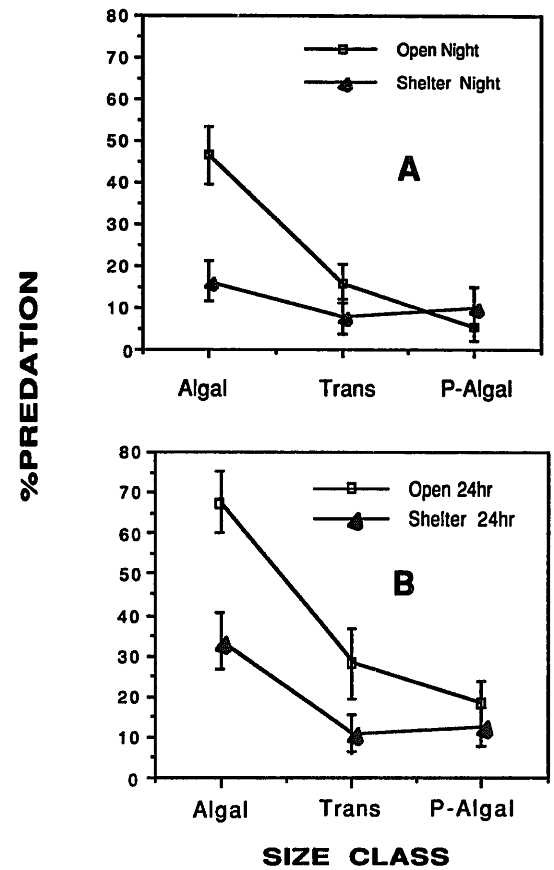

Fig. 3. Percent predation for combined sites during (A) 10-h nighttime and (B) 24-h periods. Data are represented as mean percent predation ± 1 SE for each size group in open or in shelter. Algal, algal lobsters (n = 41 open; 41 shelter); Trans, transitional lobsters (n = 46 open; 48 shelter); P-Algal, Post-algal lobsters (n = 47 open; 48 shelter).

than that of either transitional or post-algal juveniles in the open ($p < 0.001$ and $p < 0.001$, respectively) (Table IIIA). There was no significant difference in predation between transitional and post-algal groups in the open at night ($p > 0.05$).

Transitional and post-algal juveniles experienced equal predation in shelter and in the open at night ($p > 0.05$ for both classes). Algal lobsters, however, experienced significantly reduced overnight predation in algal shelter ($p < 0.001$). Therefore, the significant interaction term found in the two-factor ANOVA above was caused largely by the difference between algal shelter vs. open treatments. Post-algal lobsters in dens suffered as much predation as in the open ($p > 0.05$). However, there was no significant difference in mortality among the three groups sheltering at night ($p > 0.05$) and predation on nocturnally sheltering algal lobsters was equivalent ($p > 0.05$) to that of larger juveniles in the open.

The 24-h plots (Figs. 3B, 4) and statistical comparisons (Table IIIB) indicate substantial daytime predation on all sizes of lobsters that survived overnight in the open.

TABLE II

(A) Results of a two-way ANOVA for transformed* overnight (10 h) juvenile lobster predation data; class vs. habitat type.

Source	Sum of squares	df	Mean square	F	P
Class	0.681	2	0.340	13.53	<0.001
Habitat	0.210	1	0.210	8.35	0.006
Interaction	0.356	2	0.178	7.08	0.002
Error	1.359	54	0.025		

(B) Results of a two-way ANOVA for transformed* total (24 h) juvenile lobster predation data; class vs. habitat type.

Source	Sum of squares	df	Mean square	F	P
Class	1.333	2	0.667	11.32	<0.001
Shelter	0.518	1	0.518	8.79	0.004
Interaction	0.157	2	0.078	1.33	0.273
Error	3.179	54	0.059		

* Data are calculated mean % predation for each class vs. habitat combination transformed prior to analysis by arcsine (% predation).

Diurnal sheltering decreased predation risk for all three groups but especially the algal and transitional phases. Of special interest is the high daytime mortality on algal stages tethered in algae indicating that this size class is at higher relative risk than their older cohorts. Statistical comparisons of relevant pairwise comparisons for 24-h data are identical to those for nighttime data (Table IIB). Transitional juveniles tethered in shelter experience an apparently lower predation risk when compared to those tethered in the open, however this difference is not significant ($p > 0.05$) for 24-h data.

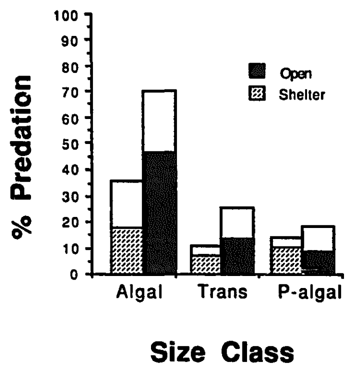

Fig. 4. Total predation on different size classes of juvenile lobsters over a 24-h period by habitat type. Color break indicates relative amount of predation which occurred over 10-h nighttime period. Blank space above color break shows relative amount of daytime predation. n = 134 for open habitat; n = 113 for shelter habitat.

TABLE III

(A) Results of nine paired Bonferroni t tests of overnight (10 h) juvenile lobster predation data transformed* for relevant class vs. shelter comparisons.

Pairwise Comparison	df	Pairwise Mean Difference	Adjusted P**
Class 1, Open vs.			
Class 1, Shelter	54	−0.325	<0.001
Class 2, Open	54	−0.383	<0.001
Class 3, Open	54	−0.427	<0.001
Class 1, Shelter vs.			
Class 2, Shelter	54	−0.079	–
Class 3, Shelter	54	−0.058	–
Class 2, Open vs.			
Class 2, Shelter	54	−0.074	–
Class 3, Open	54	0.096	–
Class 2, Shelter vs.			
Class 3, Shelter	54	−0.021	–
Class 3, Open vs.			
Class 3, Shelter	54	0.044	–

(B) Results of nine paired Bonferroni t tests of transformed* total (24 h) juvenile lobster predation data for relevant class vs. shelter comparisons.

Pairwise Comparison	df	Pairwise Mean Difference	Adjusted P**
Class 1, Open vs.			
Class 1, Shelter	54	−0.323	0.05
Class 2, Open	54	−0.372	0.02
Class 3, Open	54	−0.461	0.001
Class 1, Shelter vs.			
Class 2, Shelter	54	−0.206	–
Class 3, Shelter	54	−0.215	–
Class 2, Open vs.			
Class 2, Shelter	54	−0.157	–
Class 3, Open	54	0.089	–
Class 2, Shelter vs.			
Class 3, Shelter	54	0.010	–
Class 3, Open vs.			
Class 3, Shelter	54	−0.078	–

* Data are calculated mean % predation for each class vs. shelter combination transformed prior to analysis by arcsine (% predation).

** Adjusted P values represent accurate tests of significance for nine a posteriori comparisons using Bonferroni method. Class 1, algal lobsters; Class 2, transitional lobsters; Class 3, post-algal lobsters; [(–), not significant].

Visual censusing showed known predators of algal lobsters to be abundant at both sites (Table IV), especially grey snappers (*Lutjanus griseus* Linnaeus), grunts (*Haemulon* spp.), and portunid crabs (*Portunus* spp.) (Herrnkind & Butler, 1986). Toadfish (*Opsanus beta* Goode et Bean), although seldom seen, were abundant based on the numbers heard "calling" during each scuba predator survey. One toadfish ≈25 cm long

TABLE IV

Average numbers of potential juvenile lobster predators by site as observed during visual scuba surveys along two 60-m tethering transects. * Known juvenile spiny lobster predators; ** often heard calling within transect region; TL, estimated total length; CW, carapace width; n = 20 scuba surveys at each site.

Potential predators	KML	KOA	Size (cm)
Teleost			
Lagodon rhomboides	1–5	> 10	8–20 TL
*Lutjanus griseus**	1–5	> 10	8–30 TL
Haemulon spp.*	6–10	> 10	8–30 TL
Gerres cinereus	1–5	1–5	8–20 TL
Halichoeres spp.	-	6–10	13–20 TL
Diplectrum formosum	< 1	-	8–20 TL
Chilomycterus schoepfi	< 1	< 1	20–26 TL
*Opsanus beta**	**	**	20–31 TL
Elasmobranch			
*Ginglymostoma cirratum**	< 1	< 1	122–132 TL
Dasyatis spp.*	< 1	< 1	76–152 TL
Crustaceans			
Portunus spp.*	1–5	1–5	5–10 CW
Menippe mercenaria	< 1	-	15–20 CW

was found on the end of a tether line formerly holding a 12 mm CL lobster that the fish had apparently attacked and swallowed. Several small octopus were also found during searches through algae. Octopus are known crustacean predators and readily eat juvenile spiny lobsters in aquaria (pers. obs.). The largest lobster predators observed were small nurse sharks which rest diurnally but forage at night.

Large, transient lobster predators (Table V) captured by net included nurse sharks (n = 5; *Ginglymostoma cirratum* Bonnaterre), bonnethead sharks (n = 13; *Sphyrna tiburo* Linnaeus) and southern stingrays (n = 5; *Dasyatis americana* Hildebrand et Schroeder) in addition to bonefish (n = 24; *Albula vulpes* Linnaeus) and a permit (*Trachinotus falcatus* Linnaeus) which feed opportunistically on a wide array of crustaceans. Most of the bonnethead sharks contained a post-algal sized lobster while the stingrays contained up to several algal and transitional juveniles.

DISCUSSION

Both ontogenetic sheltering behavior and size-influenced predation mortality over the range of juvenile spiny lobsters were tested. Newly settled lobsters obtained substantial refuge from a suite of predators by continuous sheltering, confirming the results of an earlier study in the same region by Herrnkind & Butler (1986). They reported overnight predation on tethered 7–11 mm CL juveniles ranged from 60–100% in open substratum and 15–30% in algae as compared to 50–60% (open) and 15% (algae) for

TABLE V

Crustacean predators captured in trammel nets deployed from late afternoon until midnight on 3 consecutive days reported with results of gut content analyses. CW, carapace width; TL, total length; CL, carapace length; * known juvenile lobster predator.

Crustacean predator	Size range std. length (cm)	*n*	Activity period	Crustaceans found in gut	Prey size range
Dasyatis spp.* (stingrays)	59–88 TL	5	Day + night	*Panulirus argus*	11–22 mm CL
				Gonodactylus sp.	50 mm TL
				Portunus sp.	20–30 mm CW
				Callinectes sp.	
				Penaeus sp.	17–22 mm CL
*Sphyrna tiburo** (bonnethead shark)	65–85	13	Night	*Panulirus argus*	28–46 mm CL
				Portunus sp.	20–50 mm CW
				Callinectes sp.	
				Calappa flammea	-
				Mithrax sp.	-
Albula vulpes (bonefish)	50–77	4	Day + night	*Portunus* sp.	14–42 CW
Trachinotus falcatus (permit)	89	1	Night	*Calappa flammea*	50–60 CW
				Mithrax sp.	
*Ginglymostoma cirratum** (nurse shark)	45–61	5	Day	*Penaeus* sp.	<5–20 CL
*Opsanus beta** (toadfish)	22	1	Night	*Mithrax* sp.	-

a slightly larger range of algal juveniles (5–15 mm CL) in the present study. Irrespective of local variation in predator pressure, newly settled spiny lobsters are at extreme predation risk, especially when they venture from shelter. However, until several months after settlement, they are rarely observed outside of algal cover or other complexly structured habitat (Marx & Herrnkind, 1985a; pers. obs.). This cryptic lifestyle is enabled by their ability to move about the interstices and effectively hunt abundant small resident prey. Using the tethering results as an indication of relative predation risk, algal juveniles foraging nocturnally in sheltered conditions suffer about the same predation risk as their older cohorts foraging in the open. Despite the refuge advantage of algal dwelling, algal phase lobsters suffered much higher mortality in shelter over 24 h than the larger juveniles, largely as a result of higher daytime predation. Normal, unrestrained algal lobsters are subject to additional predation risk when they traverse sparsely vegetated substratum while moving among algal clumps at night (Herrnkind & Butler, 1986), probably in search of better foraging habitat (Marx & Herrnkind, 1985b).

Predation risk declined markedly with increasing size, especially for lobsters in the open, until the middle of the transitional size range, after which predation mortality for both exposed and sheltered juveniles remained relatively constant across the size range tested. Lower mortality probably resulted mainly from reduced vulnerability to numerous, ubiquitous, small predators such as juvenile grey snapper, grunts and portunid crabs which simply cannot handle larger, stronger, longer-spined, harder-shelled juveniles. For example, we commonly observed tethered 20–30 mm CL lobsters quickly repulsing 15 cm TL (est.) snappers by antennal fending. Typically, the fish only investigated lobsters of that size briefly and ignored them thereafter. Night observations by Andree (1981) and ourselves showed that transitional and post-algal juveniles forage nocturnally near shelter on substrata much sparser and more exposed than the thick algal clumps of their prior algal phase. Because food is not available within the den, the older juveniles probably only shelter overnight in dens during molt (Lipcius & Herrnkind, 1982).

The tethering results suggest that at night the den is only slightly safer than the open for transitional and post-algal phases. Here, we suggest that the tether possibly made lobsters vulnerable to being trapped and cornered by slow or probing predators like octopus and crabs. Moreover, the dens did not house cohorts to fend away or block entry by a predator (Spanier & Zimmer-Faust, 1988). Lastly, because we chose the tethering dens we cannot specify their refuge quality as compared to those freely selected by a lobster. Tethered lobsters, however, readily accepted the predetermined dens in favor of other structure within their range of movement. Even allowing for some den-tethering bias, the 24-h trends suggest that during daytime the den affords comparatively greater refuge from predators than at night especially for the transitional phase (Fig. 4).

Comparative data on larger juvenile spiny lobsters, 35–65 mm CL, are provided by Eggleston and colleagues who also used tethering techniques to evaluate the relationship between shelter size and lobster size (Eggleston et al., 1990). In a seagrass habitat in the Mexican Caribbean, with an array of predators similar to Florida Bay, they found that 46–55 mm CL lobsters in seagrass suffered predation mortality of ≈6% per day, which was significantly greater than those in artificial dens ("casitas") 2–3% per day. Juveniles from 35–45 mm CL in casitas were preyed upon at mean rates ranging from 2–8% per day depending on the den dimensions, that is, predation on that size class varied directly with casita size. Mortality differed by shelter size but not by lobster size over the 35–65 mm CL size range, with a strong interaction between lobster size and den size. Viewing their results with ours, despite some differences in experimental questions, suggests that predation mortality throughout the spiny lobster juvenile period is a product of size, exposure time and shelter characteristics.

Juvenile spiny lobsters through the post-algal phase continually face an array of predators employing a variety of hunting techniques. Among the known lobster predators we observed, netted or knew to be present in the area, based on other studies, are numerous diurnal visual hunters that constantly scan the substratum and vegetation (snappers, grunts) or ambush (toadfish), probe (portunid crabs) or extract (stingrays) algal and smaller transitional phase juveniles from algae. At night, stingrays, toadfish, crabs, octopus, and a suite of large transient predators, including bonefish, permit, nurse sharks and bonnethead sharks, continue the assault on the entire size range of juveniles. Although none of these predators is a lobster specialist, nurse sharks and bonnetheads are reported to feed heavily on juvenile lobster in certain areas (Parsons, 1987; Cruz et al., 1986). Adjacent habitats house juvenile and adult grouper (*Epinephalus* spp., *Mycteroperca* spp.), mutton snapper (*Lutjanus analis* Cuvier) and striped burrfish (*Chilomycterus schoepfi* Walbaum) which also eat small crustaceans.

Juvenile lobsters appear to deal with omnipresent predation risk by remaining unexposed during the smallest, most vulnerable benthic stages. Heavily vegetated habitat is known to effectively protect small crustaceans generally and also juvenile blue crabs which, at larger sizes, occupy more open habitat (Wilson et al., 1987, in press). The postsettlement phases of other spiny and clawed lobsters reside in vegetated or highly protective refugia. Early juvenile Australian rock lobster *Panulirus cygnus* live in rock crevices and forage among dense seagrass that overgrows their settlement habitat (Jernakoff, 1990). This species, like *P. argus*, also confronts a wide array of predaceous fish (Howard, 1988). American lobsters *Homarus americanus* settle and spend their initial benthic life in rock rubble interstices or mud tunnels, the former providing the best predation refuge in laboratory studies (Barshaw & Lavalli, 1988).

Juvenile *P. argus* shift from full-time algal dwelling to diurnal denning at a size which renders them much less vulnerable to smaller lobster predators. It is tempting to suggest that this is the ontogenetic antipredation strategy. On the other hand, our observations suggest that lobsters over 15 mm CL become too large to effectively move about or forage within dense algae. They should hypothetically continue to benefit from avoiding predators by algal dwelling but may be obliged to increase exposure to obtain adequate food. At this stage they often continue to seek solitary daytime refuge under algal clumps (Marx & Herrnkind, 1985a), whereas post-algal stage lobsters do so infrequently. The latter typically reside under sponges or in rock crevices often in the company of conspecifics, which theoretically further enhance protection by combined defenses (Spanier & Zimmer-Faust, 1988). Increased size (and size-dependent features, e.g., spine size, antenna length, exoskeleton rigidity, strength) alone provides much reduced predation risk, as indicated by the comparatively lower level of predation on post-algal (vs. algal) juveniles tethered in the open. However, tethering probably increases actual predation levels sufficiently that such distinctions as effectiveness of nocturnal vs. diurnal sheltering cannot be readily made. The present study also does not effectively address the relative antipredation value of such potentially important factors as group defense (mostly for post-algal and older juveniles), ontogenetic changes in timing and duration of exposed foraging, or risk assessment in accord with predator abundance, all of which deserve experimental evaluation.

Potential population and fishery consequences of predation on benthic crustaceans broadly were insightfully discussed by Caddy, 1986. Recognizing that apparent anti-predator specializations of juvenile stages hint at high predator vulnerability, he called for focused study. These high-vulnerability periods in early benthic life, in addition to settlement numbers per se, potentially represent bottlenecks to population growth. In the case of *Panulirus argus*, the cryptic appearance, sheltering habitat, and general behavior of the algal and transitional phases meet these criteria. Based on relative risk, as estimated by our tethering experiments, we propose that natural mortality of juveniles is highest for these phases and serves as a significant determinant of population size. In particular, the quality of predatory refuge seems paramount to ensuring maximal recruitment to larger sizes, which are much less subject to predation mortality. Preliminary large-scale field experiments indicate marked impact by refuge quality on local population size over a wide range of settlement magnitude (Butler & Herrnkind, in press). Further research potentially will clarify this picture.

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